# PendleObolSY Update Audit Report

**Oct 15, 2025**

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# Summary

This report has been prepared for PendleObolSY smart contract, to discover issues and vulnerabilities in the source code of their Smart Contract as well as any contract dependencies that were not part of an officially recognized library. A comprehensive examination has been performed, utilizing Static Analysis and Manual Review techniques.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.

# Overview

## Project Summary

| | |
|---|---|
| Project Name | **PendleObolSY** |
| Codebase | **https://github.com/DAMM-Cap/Obol-SY** |
| Commit | **aef7eee70c804f458b67cf50e8e0f9dc461038d1** |
| Language | **Solidity** |

## Audit Summary

| | |
|---|---|
| Delivery Date | **Oct 15, 2025** |
| Audit Methodology | **Static Analysis, Manual Review** |
| Total Isssues | **4** |

## [WP-M1] Inconsistency between PendleObolSY `assetInfo()` and `exchangeRate()`

**Medium**

### Issue Description

PendleObolSY token wei has a 1:1 relationship with wstObol token wei.

`PendleObolSY.assetInfo()` declares `obol` as the asset. Based on this, the exchange rate from PendleObolSY token wei to asset token wei should be the rate from wstObol token wei to `obol` token wei.

However, `PendleObolSY.exchangeRate()` returns the rate of wstObol to stObol, which is inconsistent with the above.

`exchangeRate()` should instead return the rate between `wstObol` and `Obol` .

https://github.com/DAMM-Cap/Obol-SY/blob/cfc237da35ee2ef6f9603ff5824f660fcf081d39/src/PendleObolSY.sol#L1-L120

```
@@ 1,6 @@

/// @title PendleObolSY
/// @author [DAMM Capital](https://dammcap.finance)
contract PendleObolSY is SYBase {
    using PMath for uint256;

    address public immutable obol;
@@ 14,16 @@

    constructor(string memory _name, string memory _symbol, address _obol, address _wstObol)
        SYBase(_name, _symbol, _wstObol)
    {
        obol = _obol;
@@ 22,27 @@
```

```
28      }
29
@@ 30,32 @@
33
34      /**
35       * @dev See {SYBase-_deposit}
36       *
37       *
38       * The exchange rate of stObol to shares is 1:1
39       */
40      function _deposit(address tokenIn, uint256 amountDeposited)
41          internal
42          virtual
43          override
44          returns (uint256 amountSharesOut)
45      {
46          if (tokenIn == wstObol) {
47              amountSharesOut = amountDeposited;
48          } else if (tokenIn == obol) {
49              amountSharesOut = IWstObol(wstObol).wrapUnderlying(amountDeposited);
50          }
51      }
52
@@ 53,66 @@
67
68      /**
69       * @notice Calculates and updates the exchange rate of shares to underlying
asset token
70       * @dev this is the exchange rate of wstObol to stObol
71       */
72      function exchangeRate() public view virtual override returns (uint256) {
73          return IWstObol(wstObol).previewUnwrapToFixed(1 ether);
74      }
75
@@ 76,115 @@
116
117     function assetInfo() external view returns (AssetType assetType, address
assetAddress, uint8 assetDecimals) {
118         return (AssetType.TOKEN, obol, IERC20Metadata(obol).decimals());
119     }
120 }
```

```
123   /// @notice Preview the amount of fixed liquid staking tokens that would be
      received when unwrapping.
124   /// @param _wrappedAmount The amount of wrapped tokens to unwrap.
125   /// @return The minimum amount of fixed liquid staking tokens that would be
      received.
126   /// @dev Returns a conservative estimate (1 wei less) to account for potential
      rounding in `FixedGovLst` transfer.
127   /// The actual amount received may be up to 1 wei more due to `FixedGovLst`'s
      internal rounding behavior.
128   function previewUnwrapToFixed(uint256 _wrappedAmount) public view virtual
      returns (uint256) {
129     // At worst 1 wei less than what has been requested will be returned.
130     // The preview will return the minimum amount of assets returned.
131     return _wrappedAmount - 1;
132   }
```

## Status

✓ Fixed

# [WP-L2] Name and symbol inconsistency in PendleObolSY

**Low**

## Issue Description

- OBOL token 0x0B010000b7624eb9B3DfBC279673C76E9D29D5F7
  - name: `Obol Network`
  - symbol: `OBOL`
- wstOBOL token 0x33e4A7D15de9923C680542CB10d76ea4868123fc
  - name: `Wrapped Staked Obol`
  - symbol: `wstOBOL`

PendleObolSY's name is `SY-wstOBOL` , which matches its yieldToken wstObol

However, PendleObolSY's symbol is `SY Obol Network` (likely mistakenly written as `SY ${OBOL.name}` ), when it should probably be `SY Wrapped Staked Obol`

https://github.com/DAMM-Cap/Obol-SY/blob/cfc237da35ee2ef6f9603ff5824f660fcf081d39/script/Deploy.s.sol

```
@@ 1,4 @@
5  import "@src/PendleObolSY.sol";
6
7  address constant OBOL = 0x0B010000b7624eb9B3DfBC279673C76E9D29D5F7;
8  address constant ST_OBOL = 0x6590cBBCCbE6B83eF3774Ef1904D86A7B02c2fCC;
9  address constant WST_OBOL = 0x33e4A7D15de9923C680542CB10d76ea4868123fc;
10 address constant PENDLE_PAUSE_CONTROLLER =
   0x2aD631F72fB16d91c4953A7f4260A97C2fE2f31e;
11
12 contract Deploy is Script {
13     function run() public {
14         uint256 deployerPrivateKey = vm.envUint("DEPLOYER_PK");
15         vm.startBroadcast(deployerPrivateKey);
16
17         PendleObolSY pendleObolSY = new PendleObolSY("SY Obol Network",
   "SY-wstOBOL", OBOL, WST_OBOL);
18         pendleObolSY.transferOwnership(PENDLE_PAUSE_CONTROLLER, true, false);
19
20         vm.stopBroadcast();
```

```
21
22          console.log("PendleObolSY deployed at:", address(pendleObolSY));
23      }
24  }
```

```
    @@ 1,2 @@
3
4   import "@pendle/core/StandardizedYield/SYBase.sol";
    @@ 5,6 @@
7
8   /// @title PendleObolSY
9   /// @author [DAMM Capital](https://dammcap.finance)
10  contract PendleObolSY is SYBase {
    @@ 11,16 @@
17
18      constructor(string memory _name, string memory _symbol, address _obol, address
    _wstObol)
19          SYBase(_name, _symbol, _wstObol)
20      {
    @@ 21,27 @@
28      }
29
    @@ 30,119 @@
120 }
```

## Status

✓ Fixed

# [WP-I3] Missing changes in stObol to wstObol conversion

**Informational**

## Issue Description

`PendleObolSY._deposit()` only involves obol and wstObol, not stObol. The NatSpec documentation saying ”The exchange rate of stObol to shares is 1:1” is irrelevant. This documentation might have been left over from the stObol implementation.

https://github.com/DAMM-Cap/Obol-SY/blob/cfc237da35ee2ef6f9603ff5824f660fcf081d39/src/PendleObolSY.sol#L1-L120

```
@@ 1,4 @@
5  import "./interfaces/IRstObol.sol";
6  import "./interfaces/IWstObol.sol";
7
8  /// @title PendleObolSY
9  /// @author [DAMM Capital](https://dammcap.finance)
10 contract PendleObolSY is SYBase {
11     using PMath for uint256;
12
13     address public immutable obol;
14     address public immutable rstObol;
15     address public immutable wstObol;
16     uint256 public immutable shareScaleFactor;
17
18     constructor(string memory _name, string memory _symbol, address _obol, address
   _wstObol)
19         SYBase(_name, _symbol, _wstObol)
20     {
21         obol = _obol;
22         wstObol = _wstObol;
23         rstObol = IWstObol(_wstObol).LST();
24         shareScaleFactor = IRstObol(rstObol).SHARE_SCALE_FACTOR();
25
26         /// @dev this will allow obol to be wrapped
27         _safeApproveInf(obol, wstObol);
28     }
29
30     /*///////////////////////////////////////////////////////////////
```

```
31                      DEPOSIT/REDEEM USING BASE TOKENS
32      ////////////////////////////////////////////////////////////*/
33
34      /**
35       * @dev See {SYBase-_deposit}
36       *
37       *
38       * The exchange rate of stObol to shares is 1:1
39       */
40      function _deposit(address tokenIn, uint256 amountDeposited)
41          internal
42          virtual
43          override
44          returns (uint256 amountSharesOut)
45      {
46          if (tokenIn == wstObol) {
47              amountSharesOut = amountDeposited;
48          } else if (tokenIn == obol) {
49              amountSharesOut = IWstObol(wstObol).wrapUnderlying(amountDeposited);
50          }
51      }
52
```

```
@@ 53,96 @@
```

```
97
98      function getTokensIn() public view virtual override returns (address[] memory
    res) {
99          res = new address[](2);
100         res[0] = obol;
101         res[1] = wstObol;
102     }
103
```

```
@@ 104,107 @@
```

```
108
109     function isValidTokenIn(address token) public view virtual override returns
    (bool) {
110         return token == obol || token == wstObol;
111     }
112
```

```
@@ 113,119 @@
```

```
120   }
```

## Status

✓ Fixed

# [WP-I4] Consider using `SYBaseV2`

**Informational**

## Issue Description

The current implementation inherits from `SYBase` .

Pendle has added new functions in `SYBaseV2` and removed the permit functionality.

https://github.com/DAMM-Cap/Obol-SY/blob/cfc237da35ee2ef6f9603ff5824f660fcf081d39/src/PendleObolSY.sol#L1-L120

```
@@ 1,2 @@

3
4   import "@pendle/core/StandardizedYield/SYBase.sol";
@@ 5,6 @@

7
8   /// @title PendleObolSY
9   /// @author [DAMM Capital](https://dammcap.finance)
10  contract PendleObolSY is SYBase {
@@ 11,119 @@

120 }
```

## Status

**✓ Fixed**

# Appendix

## Timeliness of content

The content contained in the report is current as of the date appearing on the report and is subject to change without notice, unless indicated otherwise by WatchPug; however, WatchPug does not guarantee or warrant the accuracy, timeliness, or completeness of any report you access using the internet or other means, and assumes no obligation to update any information following publication.

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